

Framework for Probability Samples Learning Activity

STA304 — Summer 2026

Learning Activity 6 Overview

In this activity you will apply statistical concepts from lecture 6 related to domain estimation.

Exercise

1. An SRS of 1,500 licensed boat owners in a state was sampled from a list of 400,000 names with currently licensed boats. Define two domains: domain 1 consists of the 472 licensed boat owners who own large motorboats (defined as open motorboats longer than 16 feet) and domain 2 consists of the 1,028 licensed boat owners who do not own large motorboats. The following frequency table gives the number of respondents in each domain who reported having that number of children.

Number of Children	Domain 1	Domain 2
0	76	528
1	139	189
2	166	225
3	63	76
4	19	8
5	5	2
6	3	0
8	1	0
Total	472	1028

Estimate, along with a 95% CI,

- (a) The percentage of boat owners who have children in each domain.
- (b) The average number of children per household among boat owners in domain 1.

First write the equations with all the numbers subbed in, then use R or a calculator to get the final answer while avoiding rounding.

2. The population count of licensed boat drivers who own a large boat is unknown here (N_1 is unknown using notation from class). Define estimators for the domain size and totals as:

$$\hat{N}_h = \frac{n_h}{n} N$$

$$\hat{t}_{yd}^* = \hat{N}_d \bar{y}_d$$

- (a) Estimate the total number of children who live in households where someone owns a large motorboat.

- (b) Let u_i be defined such that $u_i = y_i$ if $i \in \mathcal{U}_d$ and 0 otherwise. Show that

$$\hat{t}_{yd}^* = N\bar{u}_{\mathcal{S}}$$

where $\bar{u}_{\mathcal{S}}$ is the sample mean of u .

- (c) Justify that $\bar{u}_{\mathcal{S}}$ is sample mean estimator under SRS and write an expression for its estimated variance.
- (d) Write an expression for the standard error of $\hat{t}_{yd}^*$.
- (e) Discuss whether you **have enough information** to calculate the standard error you found in (d).